

15.438 Fixed Income: HW5

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1 BB-rated bond pricing

Setup

We price a 5-year BB bond with face $F = \$1,000$, annual coupon rate $c = 6.5\%$, required expected return $r = 5.375\%$. Conditional default probabilities are $p_1 = 25\%$ and $p_t = 5\%$ for $t = 2, \dots, 5$, with recovery rates $R_1 = 29\%$ and $R_t = 40\%$ for $t = 2, \dots, 5$ (applied to face value, as in `default.xls`).

Let $S(t) = \prod_{i=1}^t (1 - p_i)$ denote cumulative survival through year t , with $S(0) = 1$. The probability of defaulting in year t is $S(t-1)p_t$, and the probability of surviving year t is $S(t-1)(1 - p_t) = S(t)$.

Pricing equation

Conditional on surviving to the start of year t , the bond pays the coupon cF if it survives the year (plus principal F at $t = T$), or the recovery $R_t F$ if it defaults. The price is the present value of expected cash flows:

$$P = \sum_{t=1}^T \frac{1}{(1+r)^t} \left[S(t-1) \left((1-p_t)(cF + F \mathbf{1}_{t=T}) + p_t R_t F \right) \right] \quad (1)$$

The yield to maturity y is the rate that discounts the *promised* cash flows back to this price:

$$P = \sum_{t=1}^{T-1} \frac{cF}{(1+y)^t} + \frac{cF + F}{(1+y)^T}. \quad (2)$$

Computation

Survival probabilities and per-year expected cash flows are tabulated below.

t	p_t	R_t	$S(t-1)$	$P(\text{default in } t)$	$E[CF_t]$	PV_t
1	0.2500	0.29	1.000000	0.250000	\$121.25	\$115.07
2	0.0500	0.40	0.750000	0.037500	\$ 61.31	\$ 55.22
3	0.0500	0.40	0.712500	0.035625	\$ 58.25	\$ 49.78
4	0.0500	0.40	0.676875	0.033844	\$ 55.33	\$ 44.88
5	0.0500	0.40	0.643031	0.032152	\$663.45	\$510.64
Price (sum of PVs):						\$775.59

Results

Price per \$1,000 face	\$775.59
Yield to maturity	12.86%
Credit spread (YTM – required return)	7.48%

The wedge between the 12.86% YTM and the 5.375% required return reflects the heavy front-loaded default probability and low first-year recovery: investors need a much higher promised yield because so much of the promised cash flow is expected not to be received.

2 SpringTech binomial tree

Setup

Asset value $A_0 = \$100\text{M}$ evolves like following: each period $A_{t+1} = A_t u$ or $A_{t+1} = A_t/u$, with $u = 1.10$ in parts (a) to (c) and $u = 1.15$ in part (d). Risk-neutral probability of an up-move $q = 0.55$; risk-free rate $r = 3\%$. Zero-coupon debt has face $F = \$85\text{M}$, $T = 4$. Debt holders receive F if $A_T \geq F$, otherwise $0.80 A_T$.

Let k denote the number of up-moves over 4 periods. Then $A_T = A_0 u^k (1/u)^{T-k}$ and the risk-neutral probability is $\binom{T}{k} q^k (1-q)^{T-k}$.

(a) Debt price and YTM with $u = 1.10$

Terminal nodes:

k	A_T (\$M)	Default?	Debt payoff (\$M)	RN prob
0	68.30	yes	54.64	0.04101
1	82.64	yes	66.12	0.20048
2	100.00	no	85.00	0.36754
3	121.00	no	85.00	0.29948
4	146.41	no	85.00	0.09151

The expected terminal payoff under $\mathbb{Q}$ is

$$\begin{aligned}
 E^{\mathbb{Q}}[\text{payoff}_T] &= \sum_{k=0}^4 \binom{4}{k} q^k (1-q)^{4-k} \text{payoff}_k \\
 &= 0.04101 \cdot 54.64 + 0.20048 \cdot 66.12 + 0.36754 \cdot 85 \\
 &\quad + 0.29948 \cdot 85 + 0.09151 \cdot 85 \\
 &= \$79.97\text{M},
 \end{aligned}$$

so

$$P_{\text{debt}} = \frac{79.97}{1.03^4} = \$71.05\text{M}, \quad y = \left(\frac{85}{71.05} \right)^{1/4} - 1 = 4.58\%.$$

(b) Risk-neutral default probability and expected recovery

Default occurs when $A_T < F$, i.e. at $k = 0$ or $k = 1$:

$$\mathbb{Q}(\text{default}) = 0.04101 + 0.20048 = 24.15\%.$$

The expected recovery rate (% of face), conditional on default, is

$$E^{\mathbb{Q}}[\text{recovery} \mid \text{def}] = \frac{0.04101 \cdot 54.64 + 0.20048 \cdot 66.12}{0.24148 \cdot 85} = 75.49\%.$$

(c) 5% VaR over 4 years

Per the class definition, VaR is the worst expected loss at a given confidence level under physical probabilities but the problem instructs us to treat $\mathbb{Q}$ as the physical measure for this calculation.

Ranking terminal payoffs from worst to best:

Payoff (\$M)	Prob	Cum. prob
54.64	0.04101	0.04101
66.12	0.20048	0.24148
85.00	0.36754	0.60902
85.00	0.29948	0.90849
85.00	0.09151	1.00000

The 5% VaR is the smallest loss that is exceeded with probability at most 5%. Reading down the cumulative-probability column: the only outcome with tail probability $\leq 5\%$ is the $k = 0$ node ($P = 4.10\%$). The next-worst outcome ($k = 1$) has cumulative probability 24.15%, far above the 5% limit, so the 5% VaR is bracketed inside the worst node.

$$\text{VaR}_{5\%} \approx P_{\text{debt}} - 54.64 = 71.05 - 54.64 = \$16.41\text{M}$$

The discreteness of a 4-step tree makes this estimate coarse, a finer tree would push the 5% cutoff somewhere between the $k = 0$ and $k = 1$ outcomes.

(d) Sensitivity to higher asset volatility ($u = 1.15$)

Recomputing the entire tree with $u = 1.15$:

k	A_T (\$M)	Default?	Debt payoff (\$M)	RN prob
0	57.18	yes	45.74	0.04101
1	75.61	yes	60.49	0.20048
2	100.00	no	85.00	0.36754
3	132.25	no	85.00	0.29948
4	174.90	no	85.00	0.09151

	$u = 1.10$	$u = 1.15$
Debt price	\$71.05M	\$69.73M
YTM	4.58%	5.08%
$\mathbb{Q}(\text{default})$	24.15%	24.15%
$E^{\mathbb{Q}}[\text{rec} \mid \text{def}]$	75.49%	68.22%
$\text{VaR}_{5\%}$ (loss vs price)	\$16.41M	\$23.99M

Interpretation. Higher asset volatility leaves $\mathbb{Q}(\text{default})$ unchanged in this binomial setup. Indeed, the default probability only depends on the number of terminal nodes that lie below F , which is only determined by the structure of the tree, not its scale, since the same $k = 0, 1$ states still trigger default when u rises from 1.10 to 1.15. What changes is the *severity*: the distressed nodes move further below F , dragging the expected recovery from 75.49% down to 68.22% of face value. The debt therefore loses value, the YTM widens by about 50bp, and the left-tail loss roughly doubles. This underlines the fact that volatility hurts debt holders even when default frequency does not move, because they bear loss-given-default risk.

Question 3: Risky Bond Portfolio, CDS, and CDO

Setup. Face value \$10M, coupon 7.625%, 1-year maturity (annual). Promised payment per \$1M face: $1,000,000 \times 1.07625 = \$1,076,250$. Rates: Treasury 4.385%, AAA yield 4.8%.

Default State	Recovery Rate	Probability
No defaults	100%	68%
Mild default	85%	27%
Severe default	30%	5%

Expected payoff per \$1M face:

$$\begin{aligned}
 E[\text{payoff}] &= 0.68(1.0) + 0.27(0.85) + 0.05(0.30) = 0.9250 \text{ (as fraction of promised)} \\
 &= 0.9250 \times \$1,076,250 = \$994,981.25
 \end{aligned}$$

Part (a): YTM and Current Price

Investors require an expected return of 5.75%, so the current price must satisfy:

$$P \times (1 + E[r]) = E[\text{payoff}] \implies P = \frac{\$994,981.25}{1.0575} \approx \$940,869$$

For the full \$10M face portfolio: $P_{\text{portfolio}} \approx \$9,408,694$.

The YTM solves $P \times (1 + \text{YTM}) = \text{promised payment}$:

$$\text{YTM} = \frac{\$1,076,250}{\$940,869} - 1 \approx \boxed{14.39\%}$$

The spread over Treasuries is $14.39\% - 4.385\% = 10.0\%$, reflecting default risk.

Part (b): CDS Writer Payoffs (Notional \$1M)

The CDS writer receives a premium of $10.25\% \times \$1,000,000 = \$102,500$ at year-end (always). In a default state, the writer pays the insured the shortfall: $(1 - \text{RR}) \times \$1,076,250$.

Default State	Recovery	Prob.	Shortfall Paid	Net Payoff to CDS Writer
No defaults	100%	68%	\$0	+\$102,500
Mild default	85%	27%	$0.15 \times \$1,076,250 = \$161,438$	-\$58,938
Severe default	30%	5%	$0.70 \times \$1,076,250 = \$753,375$	-\$650,875

Part (c): Replicating Strategy Payoffs

Strategy: Buy \$1M face of risky bonds at market price \$940,869; short \$940,869 PV of 1-year Treasuries (obligation at year-end: $940,869 \times 1.04385 = \$981,153$).

State	Prob.	Short Treasury	Long Risky Portfolio	Net Payoff
No defaults	68%	−\$981,153	$1.0 \times \$1,076,250 = \$1,076,250$	+\$95,097
Mild default	27%	−\$981,153	$0.85 \times \$1,076,250 = \$914,813$	−\$66,340
Severe default	5%	−\$981,153	$0.30 \times \$1,076,250 = \$322,875$	−\$658,278

Part (d): Arbitrage via CDS vs. Replicating Portfolio

Comparing the CDS writer's payoffs (part b) to the replicating portfolio (part c), the CDS *dominates* in every state:

State	CDS Net	Replicating Net	CDS Premium
No defaults	+\$102,500	+\$95,097	+\$7,403
Mild default	−\$58,938	−\$66,340	+\$7,403
Severe default	−\$650,875	−\$658,278	+\$7,403

The CDS premium of 10.25% is too high relative to the fair value implied by market prices. To lock in a **riskless profit of \$7,403 per \$1M** in every state:

1. **Sell** the CDS (collect \$102,500 premium, bear default losses).
2. **Hedge** by taking the opposite of the replicating strategy: *short* \$1M face of the risky bond and *long* an equal PV of 1-year Treasuries.

The net position is zero cost today and earns \$7,403 with certainty regardless of default outcome.

Part (e): CDO Tranches (Extra Credit)

The portfolio is restructured into a **senior tranche** (AAA, absolute priority) and a **junior tranche**.

For the senior tranche to carry negligible credit risk, it must be fully repaid even in the *worst case* (severe default, 30% recovery):

$$\text{Max senior face value} = 0.30 \times \$10,762,500 = \boxed{\$3,228,750}$$

Senior tranche price: Discount at the AAA yield of 4.8% (since it now carries AAA risk):

$$P_{\text{senior}} = \frac{\$3,228,750}{1.048} \approx \$3,081,822$$

Junior tranche price: The junior tranche absorbs all residual cash flows. Its price equals the total portfolio price minus the senior tranche price:

$$P_{\text{junior}} = \$9,408,694 - \$3,081,822 \approx \$6,326,872$$

The junior tranche, despite being a large portion of the portfolio by price, bears all first-loss risk and would be discounted at a yield well above the 14.39% portfolio YTM to compensate.